FIN 2020, Midterm Exam

9:00 AM-11:00 AM, Oct 16, 2022

Name (print clearly): _____

Student ID: _____

Instructions

- This exam has four questions; before you begin, please check to make sure that your copy has all four questions.

- Provide ***convincing*** proof of all answers and be ***concise*** in your answers. If you show the steps that led to your results, I can award partial credit for the correct approach even if your final answers are slightly off.

- Please read the following statement and sign below acknowledging that you understand the statement and promise to abide by it and please turn it in with your exam.

  Statement: I pledge on my honor that I have not given or received any unauthorized assistance on this examination. By signing below, you indicate your compliance with the above guidelines, as required by the academic code of honor.

Signature: _____

1. (25 points) True, False or it depends? Briefly explain your answers.

   (a) Expected utility is invariant under any monotonic increasing transformation.

   (b) A risk neutral investor is indifferent between accepting and rejecting a fair odd. (A fair odd is a betting on a random event with 0 expected gain or loss.)

   (c) Stock A has a higher Sharpe Ratio than stock B. Then all risk averse investors who prefer more to less would prefer stock A to B.

   (d) For any risk asset, investor X's certainty equivalent of the asset is always lower than investor Y's certainty equivalent of the same asset. Then investor X must be more risk averse than investor Y.

   (e) Asset A first order stochastic dominates (FOSD) asset B, then asset A must state-by-state dominates asset B.

2. (25 points) Consider an investor whose initial wealth is \$10. His bernoulli utility is

$$u(x) = \begin{cases} x & \text{for} \quad x \leq 10 \\ 0.5x + 5 & \text{for} \quad x \geq 10 \end{cases}$$

This investor is offered two lotteries $L_1 = (-\$3, \$3, 0.5)$ and $L_2 = (-\$6, \$6, 0.5)$. First judge whether the following statements are right or wrong. If the statement is right, please provide a brief explanation. If the statement is wrong, please give the correct statement.

(a) This investor is risk neutral.

(b) If L_1's reward is \$0.7, then the investor is willing to participate in L_1.

(c) In order to persuade the investor to participate in L_2, L_2's reward must be higher than L_1's reward.

(d) If the investor's initial wealth is increased to 15, then this investor is indifferent between L_1 and L_2.

(e) If the investor's initial wealth is increased to 20, then this investor is indifferent between L_1 and L_2.

3. (25 points) Consider an investor with initial wealth $Y_0 = 10$, who can choose to invest in a risky asset. The random payoff $\tilde{d}$ of the risky asset can be described by a lottery (d^G, d^B, π), where $d^G = 3$, $d^B = 0$, and $\pi = \frac{1}{2}$. Suppose this investor has a von Neumann-Morgenster expected utility function, with a Bernoulli utility function of the form

$$u(c) = \frac{c^{1-\gamma} - 1}{1 - \gamma}.$$

Recall that the certainty equivalent $\mathrm{CE}(\tilde{d})$ of the risky asset is the maximum riskless payoff that the investor is willing to exchange for the risky asset, and the risk premium for the risky asset $\Psi(\tilde{d})$ is the difference between its expected payoff and its certainty equivalent, i.e., $\Psi(\tilde{d}) = \mathrm{E}[\tilde{d}] - \mathrm{CE}(\tilde{d})$.

(1) Assuming that the investor's constant coefficient of relative risk aversion is $\gamma = 1/2$, then what is certainty equivalent for the risky asset?

(2) Assuming that the investor's constant coefficient of relative risk aversion is $\gamma = 2$, then what is certainty equivalent for the risky asset?

(3) What is the risk premium for the risky asset when $\gamma = 1/2$? What is the risk premium for the risky asset when $\gamma = 2$? Which one is bigger?

4. *(25 points)* Consider a risk-averse investor with vN-M expected utility who ha initial wealth Y and is offered a relative bet: win kY with probability π and lose kY with probability $1-\pi$. In class, we derived the Taylor approximation

$$\pi^* \approx \frac{1}{2} + \frac{1}{4} k R_R(Y)$$

for the probability of winning π^* that makes the investor indifferent between accepting and rejecting the bet, where $R_R(Y)$ is his or her coefficient of relative risk aversion and k is the size of the bet measured as a fraction of wealth. Use this approximation to answer the following questions:

(1) Suppose that the investor's initial wealth is $Y = 100,000$ and that he or she is indifferent between the bet: win 2000 with probability 0.55 and lose 2000 with probability 0.45. What is this investor's coefficient of relative risk aversion estimated by the Taylor approximation.

(2) Suppose the investor described in part (1), above, has preference over risky alternatives that are described by a vN-M expected function with Bernoulli utility function

$$u(Y) = \frac{Y^{1-\gamma} - 1}{1-\gamma}$$

over monetary values Y received in any particular state of the world. Given your answer to part (1), above, what value would you assign to the parameter γ that appears in this Bernoulli utility function?

(3) Suppose that this same investor was offered a different bet: win 4000 with probability 0.55 and lose 4000 with probability 0.45. Given his or her coefficient of relative risk aversion, which you calculated in part (1), above, would the investor accept or reject this bet?